

Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Centre Number		Candidate Number	

Pearson Edexcel Level 3 GCE

Friday 6 June 2025

Afternoon (Time: 1 hour 30 minutes)

Paper reference **9FM0/3C**

Further Mathematics

Advanced

PAPER 3C: Further Mechanics 1

You must have:
Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

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- You should show sufficient working to make your methods clear.
- Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
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Advice

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Turn over ►

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1. A particle A has mass $4m$ and a particle B has mass $3m$. The particles are moving along the same straight line on a smooth horizontal table. The particles are moving in opposite directions towards each other when they collide directly.

As a result of the collision, the direction of motion of each particle is reversed.

Immediately **before** the collision, the speed of A is u and the speed of B is ku , where k is a constant.

Immediately **after** the collision, the speed of A is $2v$ and the speed of B is $3v$.

The magnitude of the impulse received by A in the collision is $20mv$.

- (a) Find u in terms of v only.

(3)

- (b) Find the exact value of k .

(3)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 6 marks)



2. A van of mass 900 kg is moving along a straight horizontal road.

When the speed of the van is $v\text{ ms}^{-1}$, the resistance to the motion of the van is modelled as a force of magnitude $50v\text{ N}$.

The engine of the van is working at a constant rate of 12 kW .

At the instant when the speed of the van is $V\text{ ms}^{-1}$, the acceleration of the van is 0.2 ms^{-2}

Using the model,

- (a) find the value of V .

(4)

Later on, the van is moving up a straight road that is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{12}$. When the speed of the van is $v\text{ ms}^{-1}$, the resistance to the motion of the van from non-gravitational forces is modelled as a force of magnitude $50v\text{ N}$.

When the van has constant speed $U\text{ ms}^{-1}$, the engine of the van is working at a constant rate of 15 kW .

Using the model,

- (b) find the value of U .

(4)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. A particle P of mass 0.4 kg is moving with velocity $7\mathbf{i} \text{ ms}^{-1}$ when it receives an impulse of magnitude $\sqrt{1.6} \text{ N s}$.

The velocity of P immediately after it receives the impulse is $\lambda(2\mathbf{i} + \mathbf{j}) \text{ ms}^{-1}$, where λ is a constant.

Find the two possible values of λ

(6)



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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 6 marks)



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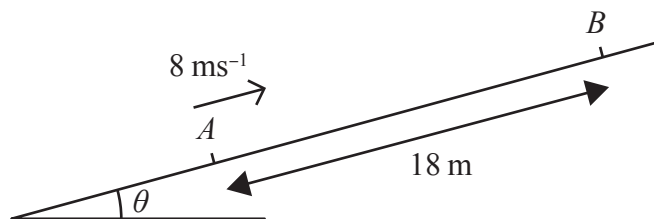


Figure 1

A rough straight ramp is fixed to horizontal ground. The ramp is inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{14}$. The points A and B are on a line of greatest slope of the ramp with $AB = 18$ m and B above A , as shown in Figure 1.

A package of mass 0.5 kg is projected up the ramp from A with speed 8 m s^{-1} and comes to instantaneous rest at B .

The work done against friction as the package moves from A to B is W joules.

The package is modelled as a particle and air resistance is ignored.

- (a) Use the work-energy principle to show that $W = 9.7$ (3)

The coefficient of friction between the package and the ramp is μ

- (b) Find the value of μ (4)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 7 marks)



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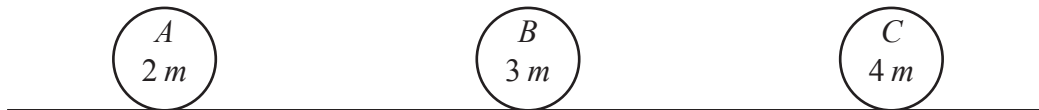


Figure 2

Three small balls, A , B and C , have masses $2m$, $3m$ and $4m$ respectively.

The balls are initially at rest in a straight line on a smooth horizontal surface, with B between A and C , as shown in Figure 2.

Ball A is projected towards B with speed $5u$ and A and B collide directly. Immediately after the collision, the speed of B is $3u$

The balls are modelled as uniform spheres with equal radii.

- (a) Find, in terms of u , the speed of A immediately after the collision with B . (3)

- (b) Find the coefficient of restitution between A and B .
- (3)**

After the collision between A and B , ball C is projected towards B with speed u . Balls B and C collide directly.

The coefficient of restitution between B and C is f

Given that there is a second direct collision between A and B

- (c) find the complete range of possible values of f (7)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 13 marks)



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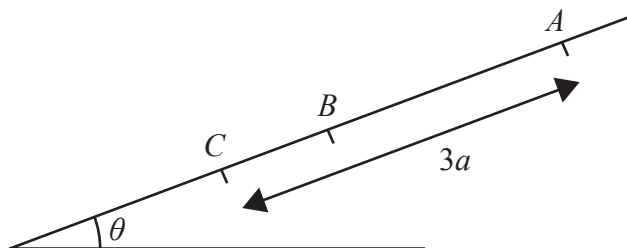


Figure 3

A smooth plane is inclined at an angle θ to horizontal ground, where $\sin \theta = \frac{1}{3}$

The points A, B and C lie on a line of greatest slope of the plane, with B between A and C , with A above B and $AC = 3a$, as shown in Figure 3.

A light elastic string of natural length $2a$ has one end attached to the fixed point A . The other end of the string is attached to a parcel P of mass m .

The modulus of elasticity of the string is $\frac{8}{3}mg$

The parcel rests in equilibrium on the plane at the point B .

The parcel is modelled as a particle.

- (a) Show that $AB = \frac{9}{4}a$ (4)

The parcel is now held at C and released from rest.

- (b) Find the elastic potential energy lost by the string when P moves from C to B . (3)

- (c) Use the principle of conservation of mechanical energy to find the speed of P at the instant it passes B . (3)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 10 marks)



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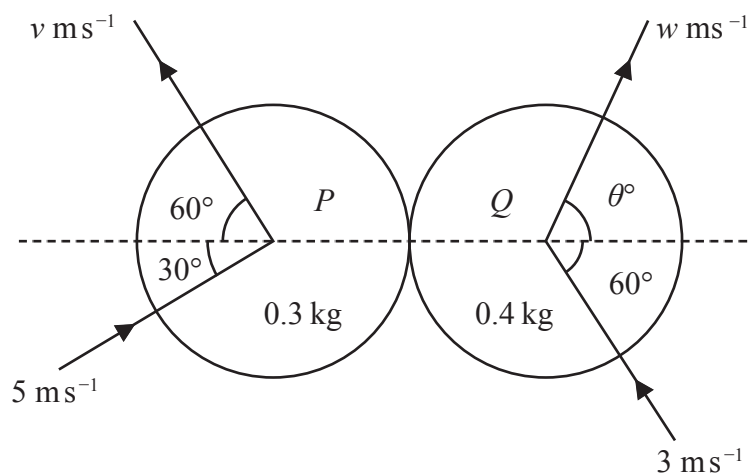


Figure 4

Two uniform spheres P and Q have equal radii. The mass of P is 0.3 kg and the mass of Q is 0.4 kg . The spheres are moving on a smooth horizontal plane when they collide obliquely.

Immediately **before** the collision,

- P is moving with speed 5 ms^{-1} at 30° to the line of centres of the spheres
- Q is moving with speed 3 ms^{-1} at 60° to the line of centres of the spheres

Immediately **after** the collision,

- P is moving with speed $v \text{ ms}^{-1}$ at 60° to the line of centres of the spheres
- Q is moving with speed $w \text{ ms}^{-1}$ at θ° to the line of centres of the spheres

as in the plan view shown in Figure 4.

(a) Show that $v = \frac{5\sqrt{3}}{3}$ (3)

- (b) Find (i) the value of w
 (ii) the value of θ (7)

- (c) Find the coefficient of restitution between P and Q . (3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 13 marks)



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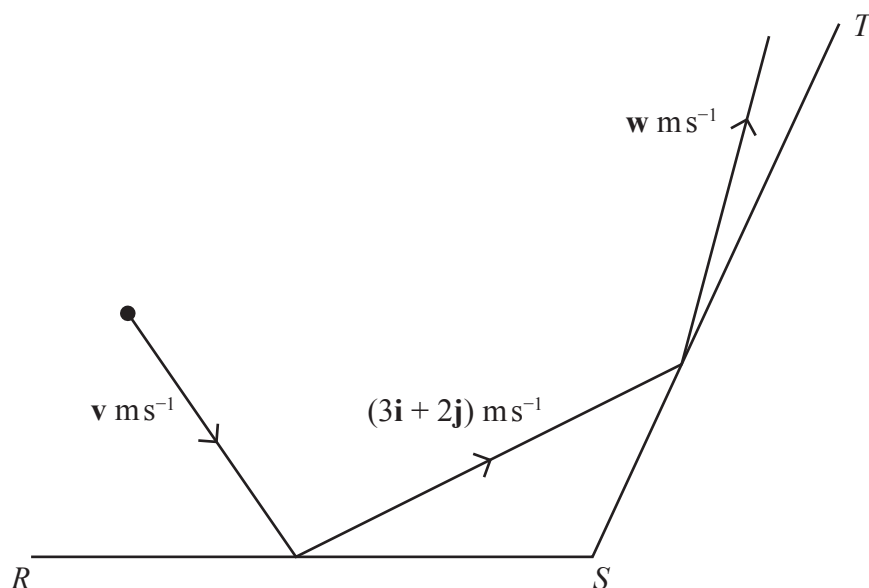


Figure 5

Figure 5 represents the plan view of part of a smooth horizontal floor, where RS and ST are smooth fixed vertical walls.

The vector $\overrightarrow{RS}$ is in the direction of the vector $\mathbf{i}$.

The vector $\overrightarrow{ST}$ is in the direction of the vector $(3\mathbf{i} + 4\mathbf{j})$.

A small ball B of mass 0.25 kg is projected across the floor towards RS .

Immediately before the impact with RS , the velocity of B is $\mathbf{v} \text{ ms}^{-1}$

Immediately after the impact with RS , the velocity of B is $(3\mathbf{i} + 2\mathbf{j}) \text{ ms}^{-1}$

The coefficient of restitution between B and RS is $\frac{1}{3}$

The ball is modelled as a particle.

(a) Show that the kinetic energy lost by B in the impact with RS is 4 J .

(6)

Immediately after the impact with ST , the velocity of B is $\mathbf{w} \text{ ms}^{-1}$

Given that the coefficient of restitution between B and ST is $\frac{1}{3}$

(b) find $\mathbf{w}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

(6)



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Question 8 continued

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Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 12 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Friday 7 June 2024

Afternoon (Time: 1 hour 30 minutes) **Paper reference** **9FM0/3C**

Further Mathematics
Advanced
PAPER 3C: Further Mechanics 1

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 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

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Turn over ►

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1. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal perpendicular unit vectors.]

A particle A has mass 3 kg and a particle B has mass 2 kg.

The particles are moving on a smooth horizontal plane when they collide directly.

Immediately **before** the collision, the velocity of A is $(3\mathbf{i} - \mathbf{j})\text{ m s}^{-1}$ and the velocity of B is $(-6\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$

Immediately **after** the collision the velocity of A is $\left(-2\mathbf{i} + \frac{2}{3}\mathbf{j}\right)\text{ m s}^{-1}$

- (a) Find the total kinetic energy of the two particles **before** the collision. (3)
- (b) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the impulse exerted on A by B in the collision. (3)
- (c) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B immediately **after** the collision. (3)



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Question 1 continued

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(Total for Question 1 is 9 marks)



2. A rough plane is inclined to the horizontal at an angle θ , where $\tan \theta = \frac{3}{4}$

A particle P of mass m is at rest at a point on the plane.

The particle is projected **up** the plane with speed $\sqrt{2ag}$

The particle moves up a line of greatest slope of the plane and comes to instantaneous rest after moving a distance d .

The coefficient of friction between P and the plane is $\frac{1}{7}$

- (a) Show that the magnitude of the frictional force acting on P as it moves up the plane

is $\frac{4mg}{35}$ (3)

Air resistance is assumed to be negligible.

Using the work-energy principle,

- (b) find d in terms of a . (4)



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Question 2 continued

Handwriting practice area with horizontal lines.

(Total for Question 2 is 7 marks)



3. A car of mass 1000 kg moves in a straight line along a horizontal road at a constant speed of 72 km h^{-1}

- The resistance to the motion of the car is modelled as a constant force of magnitude 900 N

The engine of the car is working at a constant rate of $P\text{ kW}$.

Using the model,

- (a) find the value of P .

(3)

The car now travels in a straight line up a road which is inclined to the horizontal at an

angle α , where $\sin \alpha = \frac{2}{49}$

- In a refined model, the resistance to the motion of the car from non-gravitational forces is now modelled as a force of magnitude $20v$ newtons, where $v\text{ m s}^{-1}$ is the speed of the car

At the instant when the engine of the car is working at a constant rate of 30 kW and the car is moving up the road at 10 m s^{-1} , the acceleration of the car is $a\text{ m s}^{-2}$

Using the refined model,

- (b) find the value of a .

(4)

Later on, when the engine of the car is again working at a constant rate of 30 kW , the car is moving up the road at a constant speed $U\text{ m s}^{-1}$

Using the refined model,

- (c) find the value of U .

(5)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 12 marks)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Handwriting practice area with horizontal lines.

(Total for Question 4 is 15 marks)



5. A light elastic string has natural length $2a$ and modulus of elasticity $2mg$. One end of the string is attached to a fixed point A on a horizontal ceiling. The other end is attached to a particle P of mass m .

The particle P hangs in equilibrium at the point E , where $AE = 3a$.

The particle P is then projected vertically downwards from E with speed $\frac{3}{2}\sqrt{ag}$

Air resistance is assumed to be negligible.

Find the elastic energy stored in the string, when P first comes to instantaneous rest. Give your answer in the form $kmg a$, where k is a constant to be found.

(7)



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Question 5 continued

Handwriting practice area with 25 horizontal lines.

(Total for Question 5 is 7 marks)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 10 marks)



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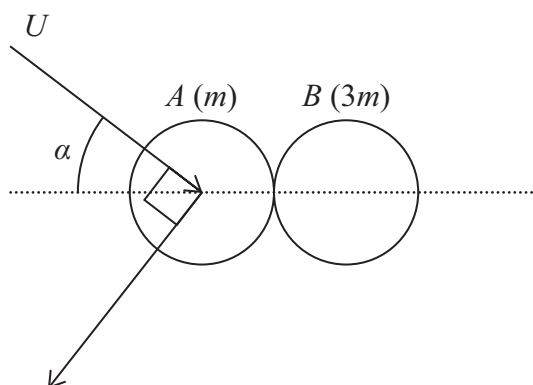


Figure 1

A smooth uniform sphere A of mass m is moving with speed U on a smooth horizontal plane. The sphere A collides obliquely with a smooth uniform sphere B of mass $3m$ which is at rest on the plane. The two spheres have the same radius.

Immediately before the collision, the direction of motion of A makes an angle α , where $0^\circ < \alpha < 90^\circ$, with the line joining the centres of the spheres.

Immediately after the collision, the direction of motion of A is **perpendicular** to its original direction, as shown in Figure 1.

The coefficient of restitution between the spheres is e .

(a) Show that the speed of B immediately after the collision is

$$\frac{1}{4}(1+e)U\cos\alpha \quad (6)$$

(b) Show that $e > \frac{1}{3}$ (4)

(c) Show that $0 < \tan\alpha \leq \frac{1}{\sqrt{2}}$ (5)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS



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Pearson Edexcel Level 3 GCE

Wednesday 14 June 2023

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Further Mathematics

Advanced

PAPER 3C: Further Mechanics 1

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Turn over ►

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1. A particle P of mass 2 kg is moving with velocity $(-4\mathbf{i} + 3\mathbf{j})\text{ m s}^{-1}$ when it receives an impulse $(-6\mathbf{i} + 42\mathbf{j})\text{ N s}$.

(a) Find the speed of P immediately after receiving the impulse.

(4)

The angle through which the direction of motion of P has been deflected by the impulse is α°

(b) Find the value of α

(2)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2. A car of mass 1000 kg moves in a straight line along a horizontal road at a constant speed $U \text{ ms}^{-1}$. The resistance to the motion of the car is a constant force of magnitude 400 N.

The engine of the car is working at a constant rate of 16 kW.

- (a) Find the value of U .

(3)

The car now pulls a trailer of mass 600 kg in a straight line along the road using a tow rope which is parallel to the direction of motion. The resistance to the motion of the car is again a constant force of magnitude 400 N. The resistance to the motion of the trailer is a constant force of magnitude 300 N.

The engine of the car is working at a constant rate of 16 kW.

The tow rope is modelled as being light and inextensible.

Using the model,

- (b) find the tension in the tow rope at the instant when the speed of the car is $\frac{20}{3} \text{ ms}^{-1}$

(5)



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Question 2 continued

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Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 8 marks)



3. A particle P of mass $2m$ is moving in a straight line with speed $3u$ on a smooth horizontal plane. It collides directly with a particle Q of mass m that is moving on the plane with speed $2u$ in the opposite direction to P .

The coefficient of restitution between P and Q is e , where $e > \frac{4}{5}$

- (a) Show that the speed of Q immediately after the collision is $\frac{(4 + 10e)u}{3}$ (6)

After the collision Q hits a smooth fixed vertical wall that is perpendicular to the direction of motion of Q . The coefficient of restitution between Q and the wall is f .

- (b) Find, **in terms of e** , the set of values of f for which there will be a second collision between P and Q . (4)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 10 marks)



4. A light elastic string has natural length $2a$ and modulus of elasticity $4mg$. One end of the elastic string is attached to a fixed point O . A particle P of mass m is attached to the other end of the elastic string. The particle P hangs freely in equilibrium at the point E , which is vertically below O .
- (a) Find the length OE .

(4)

Particle P is now pulled vertically downwards to the point A , where $OA = 4a$, and released from rest. The resistance to the motion of P is a constant force of magnitude $\frac{1}{4}mg$.

- (b) Find, in terms of a and g , the speed of P after it has moved a distance a .

(7)

Particle P is now held at O

Particle P is released from rest and reaches its maximum speed at the point B .

The resistance to the motion of P is again a constant force of magnitude $\frac{1}{4}mg$.

- (c) Find the distance OB .

(4)

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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 15 marks)



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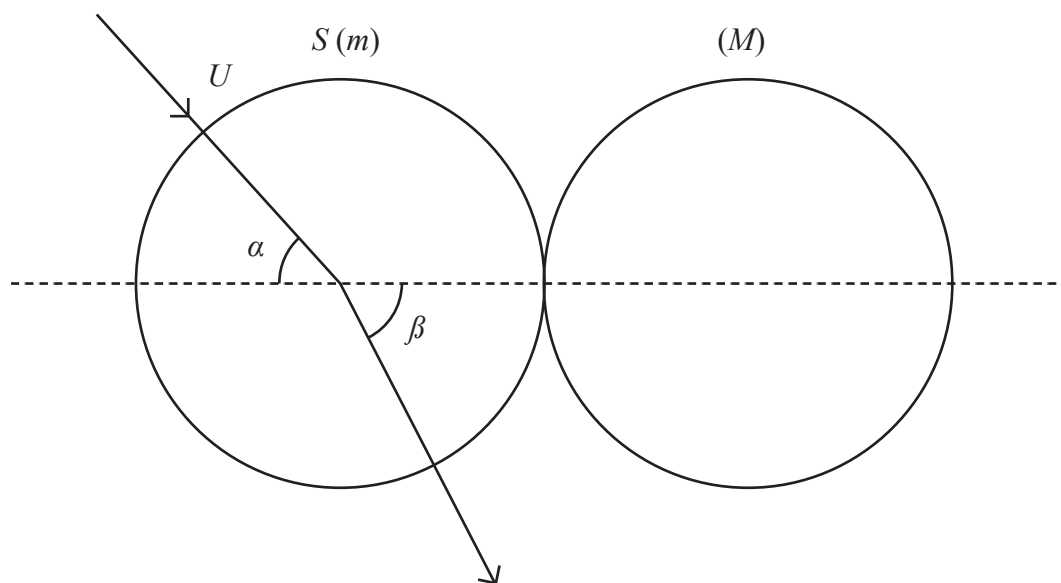


Figure 1

A smooth uniform sphere S of mass m is moving with speed U on a smooth horizontal plane. The sphere S collides obliquely with another uniform sphere of mass M which is at rest on the plane. The two spheres have the same radius.

Immediately before the collision the direction of motion of S makes an angle α , where $0 < \alpha < 90^\circ$, with the line joining the centres of the spheres.

Immediately after the collision the direction of motion of S makes an angle β with the line joining the centres of the spheres, as shown in Figure 1.

The coefficient of restitution between the spheres is e .

(a) Show that $\tan \beta = \frac{(m + M) \tan \alpha}{(m - eM)}$ (8)

Given that $m = eM$,

(b) show that the directions of motion of the two spheres immediately after the collision are perpendicular. (2)

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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 10 marks)



6. A particle P of mass m is falling vertically when it strikes a fixed smooth inclined plane. The plane is inclined to the horizontal at an angle α , where $0 < \alpha \leq 45^\circ$

At the instant immediately before the impact, the speed of P is u .

At the instant immediately after the impact, P is moving horizontally with speed v .

- (a) Show that the magnitude of the impulse exerted on the plane by P is $mu \sec \alpha$ (5)

The coefficient of restitution between P and the plane is e , where $e > 0$

- (b) Show that $v^2 = u^2(\sin^2 \alpha + e^2 \cos^2 \alpha)$ (3)

- (c) Show that the kinetic energy lost by P in the impact is

$$\frac{1}{2}mu^2(1 - e^2)\cos^2\alpha \quad (2)$$

- (d) Hence find, in terms of m , u and e **only**, the kinetic energy lost by P in the impact. (2)

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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 12 marks)



7.

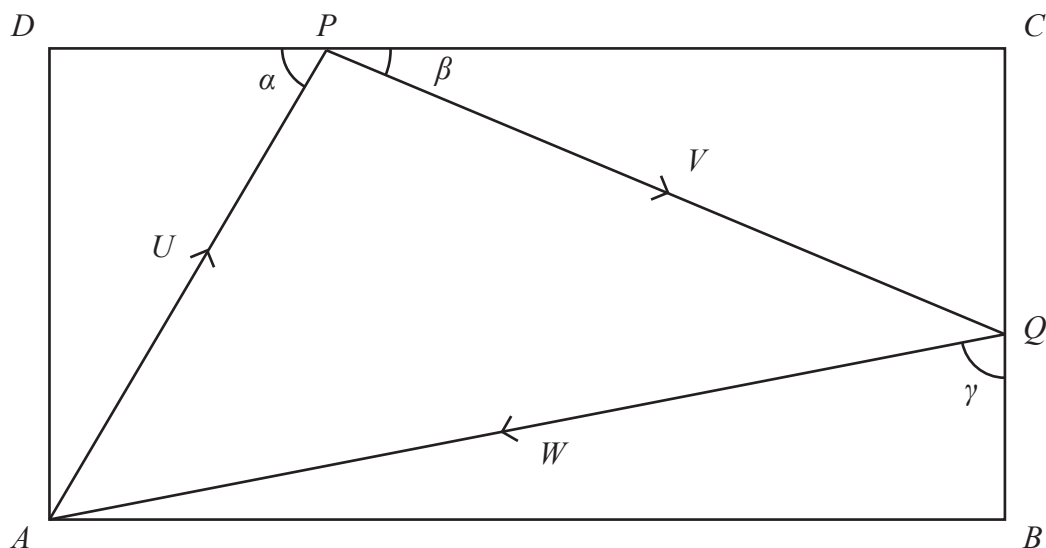


Figure 2

A small smooth snooker ball is projected from the corner A of a horizontal rectangular snooker table $ABCD$.

The ball is projected so it first hits the side DC at the point P , then hits the side CB at the point Q and then returns to A .

Angle $APD = \alpha$, Angle $QPC = \beta$, Angle $AQB = \gamma$

The ball moves along AP with speed U , along PQ with speed V and along QA with speed W , as shown in Figure 2.

The coefficient of restitution between the ball and side DC is e_1

The coefficient of restitution between the ball and side CB is e_2

The ball is modelled as a particle.

Use the model to answer all parts of this question.

(a) Show that $\tan \beta = e_1 \tan \alpha$ (4)

(b) Hence show that $e_1 \tan \alpha = e_2 \cot \gamma$ (3)

(c) By considering (angle APQ + angle AQP) or otherwise, show that it would be possible for the ball to return to A only if $e_2 > e_1$ (6)

If instead $e_1 = e_2$, the ball would **not** return to A .

Given that $e_1 = e_2$

(d) use the result from part (b) to describe the path of the ball after it hits CB at Q , explaining your answer. (1)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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(Total for Question 7 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



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Other names

Centre Number

Candidate Number

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Pearson Edexcel Level 3 GCE

Time 1 hour 30 minutes

Paper
reference

9FM0/3C

Further Mathematics

Advanced

PAPER 3C: Further Mechanics 1

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. A particle A of mass $3m$ and a particle B of mass m are moving along the same straight line on a smooth horizontal surface. The particles are moving in opposite directions towards each other when they collide directly.

Immediately before the collision, the speed of A is ku and the speed of B is u .

Immediately after the collision, the speed of A is v and the speed of B is $2v$.

The magnitude of the impulse received by B in the collision is $\frac{3}{2}mu$.

- (a) Find v in terms of u only.

(3)

- (b) Find the two possible values of k .

(5)



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Question 1 continued

Lined area for writing the answer to Question 1.



Question 1 continued

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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 8 marks)



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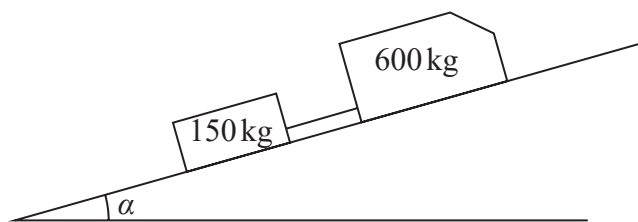


Figure 1

A van of mass 600 kg is moving up a straight road which is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{15}$. The van is towing a trailer of mass 150 kg. The van is attached to the trailer by a towbar which is parallel to the direction of motion of the van and the trailer, as shown in Figure 1.

The resistance to the motion of the van from non-gravitational forces is modelled as a constant force of magnitude 200 N.

The resistance to the motion of the trailer from non-gravitational forces is modelled as a constant force of magnitude 100 N.

The towbar is modelled as a light rod.

The engine of the van is working at a constant rate of 12 kW.

Find the tension in the towbar at the instant when the speed of the van is 9 m s^{-1}

(8)

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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

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(Total for Question 2 is 8 marks)



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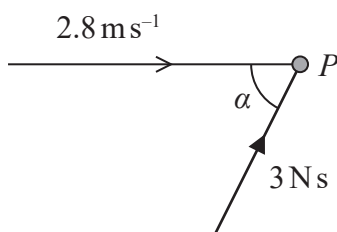


Figure 2

A particle P of mass 0.5 kg is moving in a straight line with speed 2.8 ms^{-1} when it receives an impulse of magnitude 3 N s .

The angle between the direction of motion of P immediately before receiving the impulse and the line of action of the impulse is α , where $\tan \alpha = \frac{4}{3}$, as shown in Figure 2.

Find the speed of P immediately after receiving the impulse.

(5)



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Question 3 continued

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(Total for Question 3 is 5 marks)



4.

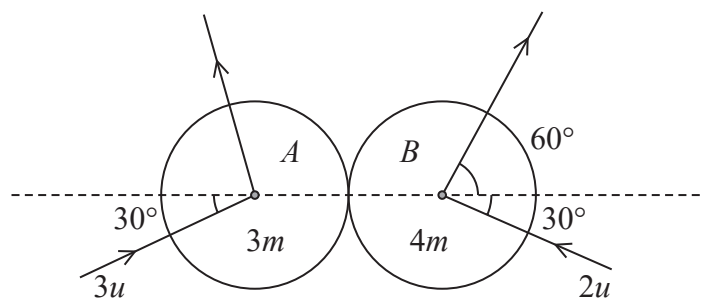


Figure 3

Two smooth uniform spheres, A and B , have equal radii. The mass of A is $3m$ and the mass of B is $4m$. The spheres are moving on a smooth horizontal plane when they collide obliquely. Immediately before they collide, A is moving with speed $3u$ at 30° to the line of centres of the spheres and B is moving with speed $2u$ at 30° to the line of centres of the spheres. The direction of motion of B is turned through an angle of 90° by the collision, as shown in Figure 3.

- (i) Find the size of the angle through which the direction of motion of A is turned as a result of the collision.
- (ii) Find, in terms of m and u , the magnitude of the impulse received by B in the collision.

(9)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



5. Two particles, P and Q , are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

The mass of P is $3m$ and the mass of Q is $4m$.

Immediately before the collision the speed of P is $2u$ and the speed of Q is u .

The coefficient of restitution between P and Q is e .

- (a) Show that the speed of Q immediately after the collision is $\frac{u}{7}(9e + 2)$

(6)

After the collision with P , particle Q collides directly with a fixed vertical wall and rebounds. The wall is perpendicular to the direction of motion of Q .

The coefficient of restitution between Q and the wall is $\frac{1}{2}$

- (b) Find the complete range of possible values of e for which there is a second collision between P and Q .

(4)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

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(Total for Question 5 is 10 marks)



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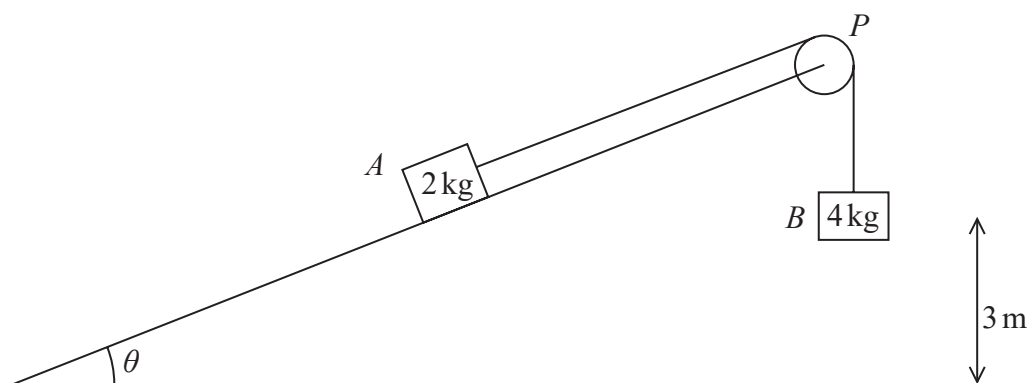


Figure 4

Two blocks, A and B , of masses 2 kg and 4 kg respectively are attached to the ends of a light inextensible string.

Initially A is held on a fixed rough plane. The plane is inclined to horizontal ground at an angle θ , where $\tan \theta = \frac{3}{4}$

The string passes over a small smooth light pulley P that is fixed at the top of the plane. The part of the string from A to P is parallel to a line of greatest slope of the plane.

Block A is held on the plane with the distance AP greater than 3 m .

Block B hangs freely below P at a distance of 3 m above the ground, as shown in Figure 4.

The coefficient of friction between A and the plane is μ

Block A is released from rest with the string taut.

By modelling the blocks as particles,

- (a) find the potential energy lost by the whole system as a result of B falling 3 m . (3)

Given that the speed of B at the instant it hits the ground is 4.5 m s^{-1} and ignoring air resistance,

- (b) use the work-energy principle to find the value of μ (6)

After B hits the ground, A continues to move up the plane but does not reach the pulley in the subsequent motion.

Block A comes to instantaneous rest after moving a total distance of $(3 + d)\text{ m}$ from its point of release.

Ignoring air resistance,

- (c) use the work-energy principle to find the value of d (4)



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Question 6 continued

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Question 6 continued

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Question 6 continued

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(Total for Question 6 is 13 marks)



7. A spring of natural length a has one end attached to a fixed point A . The other end of the spring is attached to a package P of mass m . The package P is held at rest at the point B , which is vertically below A such that $AB = 3a$.

After being released from rest at B , the package P first comes to instantaneous rest at A . Air resistance is modelled as being negligible.

By modelling the spring as being light and modelling P as a particle,

- (a) show that the modulus of elasticity of the spring is $2mg$ (5)

- (b) (i) Show that P attains its maximum speed when the extension of the spring is $\frac{1}{2}a$

- (ii) Use the principle of conservation of mechanical energy to find the maximum speed, giving your answer in terms of a and g .
- (6)

In reality, the spring is not light.

- (c) State one way in which this would affect your energy equation in part (b). (1)



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Question 7 continued

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Question 7 continued

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Question 7 continued

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(Total for Question 7 is 12 marks)



8.

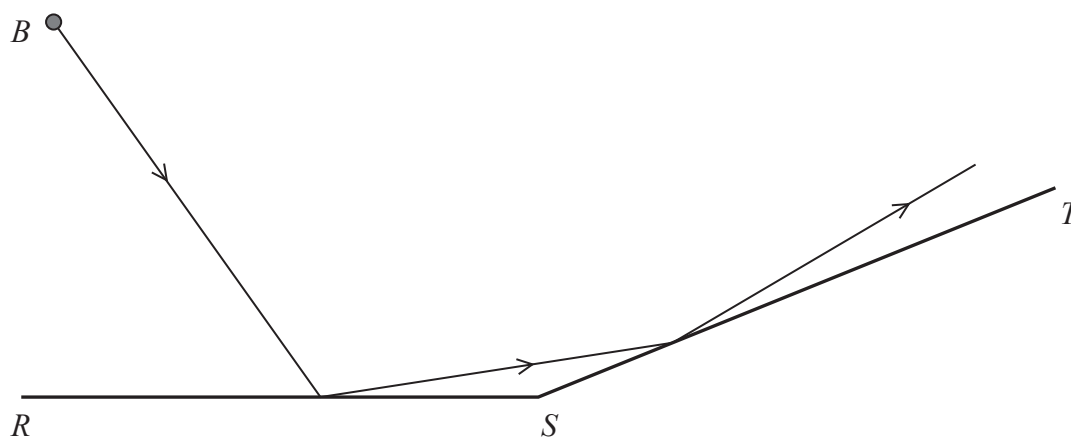


Figure 5

Figure 5 represents the plan view of part of a smooth horizontal floor, where RS and ST are smooth fixed vertical walls. The vector $\overrightarrow{RS}$ is in the direction of $\mathbf{i}$ and the vector $\overrightarrow{ST}$ is in the direction of $(2\mathbf{i} + \mathbf{j})$.

A small ball B is projected across the floor towards RS . Immediately before the impact with RS , the velocity of B is $(6\mathbf{i} - 8\mathbf{j})\text{ m s}^{-1}$. The ball bounces off RS and then hits ST .

The ball is modelled as a particle.

Given that the coefficient of restitution between B and RS is e ,

- (a) find the full range of possible values of e . (3)

It is now given that $e = \frac{1}{4}$ and that the coefficient of restitution between B and ST is $\frac{1}{2}$

- (b) Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of B immediately after its impact with ST . (7)



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Question 8 continued

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Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 10 marks)

TOTAL FOR PAPER IS 75 MARKS



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Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Time 1 hour 30 minutes	Paper reference	9FM0/3C	
Further Mathematics Advanced PAPER 3C: Further Mechanics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
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- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

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- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.
- Good luck with your examination.

Turn over ►

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1. A van of mass 900 kg is moving along a straight horizontal road.

At the instant when the speed of the van is $v \text{ m s}^{-1}$, the resistance to the motion of the van is modelled as a force of magnitude $(500 + 7v) \text{ N}$.

When the engine of the van is working at a constant rate of 18 kW, the van is moving along the road at a constant speed $V \text{ m s}^{-1}$

- (a) Find the value of V .

(5)

Later on, the van is moving up a straight road that is inclined to the horizontal at an angle θ , where $\sin \theta = \frac{1}{21}$

At the instant when the speed of the van is $v \text{ m s}^{-1}$, the resistance to the motion of the van from non-gravitational forces is modelled as a force of magnitude $(500 + 7v) \text{ N}$.

The engine of the van is again working at a constant rate of 18 kW.

- (b) Find the acceleration of the van at the instant when $v = 15$

(4)



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Question 1 continued

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Question 1 continued

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(Total for Question 1 is 9 marks)



2. Two particles, A and B , are moving in opposite directions along the same straight line on a smooth horizontal surface when they collide directly.

Particle A has mass $5m$ and particle B has mass $3m$.

The coefficient of restitution between A and B is e , where $e > 0$

Immediately **after** the collision the speed of A is v and the speed of B is $2v$.

Given that A and B are moving in the same direction after the collision,

- (a) find the set of possible values of e .

(8)

Given also that the kinetic energy of A immediately after the collision is 16% of the kinetic energy of A immediately before the collision,

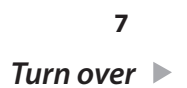
- (b) find

- (i) the value of e ,
(ii) the magnitude of the impulse received by A in the collision, giving your answer in terms of m and v .

(6)



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Question 2 continued

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Question 2 continued

Lined area for writing answers.

(Total for Question 2 is 14 marks)



3. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A smooth uniform sphere P has mass 0.3 kg . Another smooth uniform sphere Q , with the same radius as P , has mass 0.5 kg .

The spheres are moving on a smooth horizontal surface when they collide obliquely. Immediately before the collision the velocity of P is $(u\mathbf{i} + 2\mathbf{j})\text{ms}^{-1}$, where u is a positive constant, and the velocity of Q is $(-4\mathbf{i} + 3\mathbf{j})\text{ms}^{-1}$

At the instant when the spheres collide, the line joining their centres is parallel to $\mathbf{i}$.

The coefficient of restitution between P and Q is $\frac{3}{5}$

As a result of the collision, the direction of motion of P is deflected through an angle of 90° and the direction of motion of Q is deflected through an angle of α°

- (a) Find the value of u (8)
- (b) Find the value of α (5)
- (c) State how you have used the fact that P and Q have equal radii. (1)



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Question 3 continued

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Question 3 continued

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4. A particle P has mass 0.5 kg . It is moving in the xy plane with velocity $8\mathbf{i} \text{ m s}^{-1}$ when it receives an impulse $\lambda(-\mathbf{i} + \mathbf{j}) \text{ N s}$, where λ is a positive constant.

The angle between the direction of motion of P immediately before receiving the impulse and the direction of motion of P immediately after receiving the impulse is θ°

Immediately after receiving the impulse, P is moving with speed $4\sqrt{10} \text{ m s}^{-1}$

Find (i) the value of λ

(ii) the value of θ

(8)



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Question 4 continued

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(Total for Question 4 is 8 marks)



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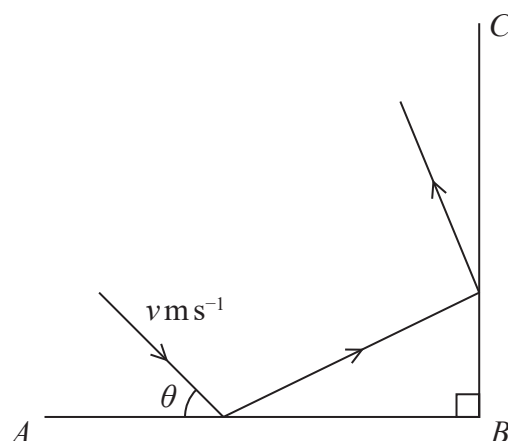


Figure 1

Figure 1 represents the plan view of part of a horizontal floor, where AB and BC represent fixed vertical walls, with AB perpendicular to BC .

A small ball is projected along the floor towards the wall AB . Immediately before hitting the wall AB the ball is moving with speed $v \text{ m s}^{-1}$ at an angle θ to AB .

The ball hits the wall AB and then hits the wall BC .

The coefficient of restitution between the ball and the wall AB is $\frac{1}{3}$.

The coefficient of restitution between the ball and the wall BC is e .

The floor and the walls are modelled as being smooth.

The ball is modelled as a particle.

The ball loses half of its kinetic energy in the impact with the wall AB .

(a) Find the exact value of $\cos \theta$.

(5)

The ball loses half of its remaining kinetic energy in the impact with the wall BC .

(b) Find the exact value of e .

(5)



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Question 5 continued

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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 10 marks)



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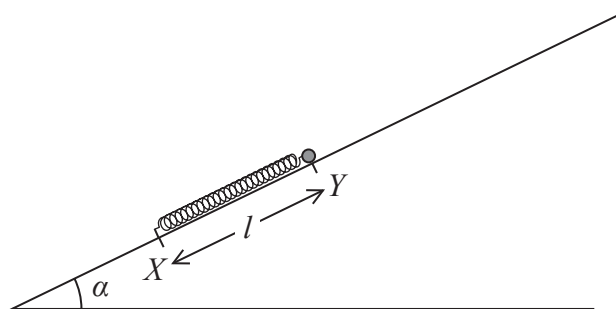


Figure 2

A light elastic spring has natural length $3l$ and modulus of elasticity $3mg$.

One end of the spring is attached to a fixed point X on a rough inclined plane.

The other end of the spring is attached to a package P of mass m .

The plane is inclined to the horizontal at an angle α where $\tan \alpha = \frac{3}{4}$

The package is initially held at the point Y on the plane, where $XY = l$. The point Y is higher than X and XY is a line of greatest slope of the plane, as shown in Figure 2.

The package is released from rest at Y and moves up the plane.

The coefficient of friction between P and the plane is $\frac{1}{3}$

By modelling P as a particle,

(a) show that the acceleration of P at the instant when P is released from rest is $\frac{17}{15}g$ (5)

(b) find, in terms of g and l , the speed of P at the instant when the spring first reaches its natural length of $3l$. (6)



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Question 6 continued

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Question 6 continued

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Question 6 continued

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(Total for Question 6 is 11 marks)



7. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

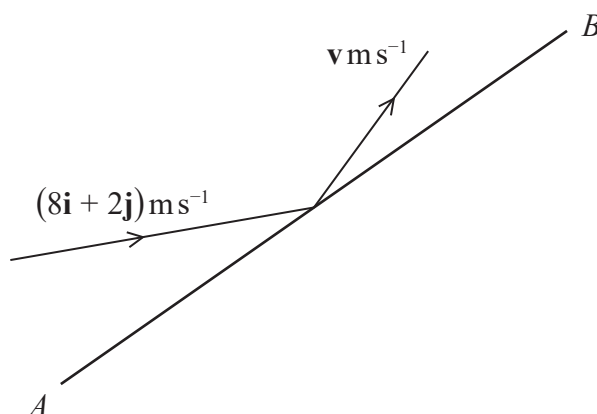


Figure 3

Figure 3 represents the plan view of part of a smooth horizontal floor, where AB is a fixed smooth vertical wall.

The direction of $\overrightarrow{AB}$ is in the direction of the vector $(\mathbf{i} + \mathbf{j})$

A small ball of mass 0.25 kg is moving on the floor when it strikes the wall AB .

Immediately before its impact with the wall AB , the velocity of the ball is $(8\mathbf{i} + 2\mathbf{j}) \text{ ms}^{-1}$

Immediately after its impact with the wall AB , the velocity of the ball is $\mathbf{v} \text{ ms}^{-1}$

The coefficient of restitution between the ball and the wall is $\frac{1}{3}$

By modelling the ball as a particle,

(a) show that $\mathbf{v} = 4\mathbf{i} + 6\mathbf{j}$ (6)

(b) Find the magnitude of the impulse received by the ball in the impact. (3)



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Question 7 continued

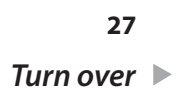
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Question 7 continued

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(Total for Question 7 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS



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Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Tuesday 20 October 2020			
Afternoon (Time: 1 hour 30 minutes)		Paper Reference 9FM0/3C	
Further Mathematics Advanced Paper 3C: Further Mechanics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

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- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. A particle P of mass 0.5 kg is moving with velocity $(4\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$ when it receives an impulse $\mathbf{J} \text{ N s}$. Immediately after receiving the impulse, P is moving with velocity $(-\mathbf{i} + 6\mathbf{j}) \text{ m s}^{-1}$.

(a) Find the magnitude of $\mathbf{J}$.

(4)

The angle between the direction of the impulse and the direction of motion of P immediately before receiving the impulse is α°

(b) Find the value of α

(3)



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Question 1 continued

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(Total for Question 1 is 7 marks)



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Question 2 continued

Lined area for writing the answer to Question 2.



Question 2 continued

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 9 marks)



3. Two particles, A and B , have masses $3m$ and $4m$ respectively. The particles are moving in the same direction along the same straight line on a smooth horizontal surface when they collide directly. Immediately before the collision the speed of A is $2u$ and the speed of B is u .

The coefficient of restitution between A and B is e .

- (a) Show that the direction of motion of each of the particles is unchanged by the collision.

(8)

After the collision with A , particle B collides directly with a third particle, C , of mass $2m$, which is at rest on the surface.

The coefficient of restitution between B and C is also e .

- (b) Show that there will be a second collision between A and B .

(6)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 14 marks)



4. [In this question, $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

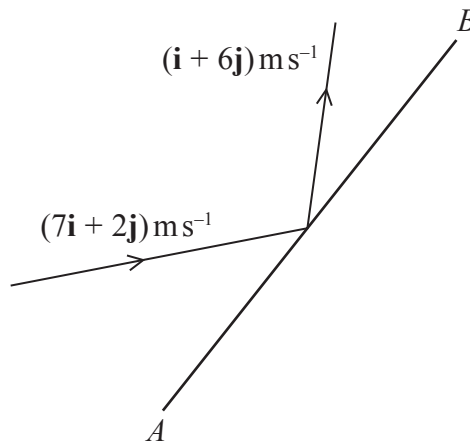


Figure 1

Figure 1 represents the plan view of part of a smooth horizontal floor, where AB represents a fixed smooth vertical wall.

A small ball of mass 0.5 kg is moving on the floor when it strikes the wall.

Immediately before the impact the velocity of the ball is $(7\mathbf{i} + 2\mathbf{j}) \text{ m s}^{-1}$.

Immediately after the impact the velocity of the ball is $(\mathbf{i} + 6\mathbf{j}) \text{ m s}^{-1}$.

The coefficient of restitution between the ball and the wall is e .

- (a) Show that AB is parallel to $(2\mathbf{i} + 3\mathbf{j})$. (4)

- (b) Find the value of e . (5)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 9 marks)



5. A smooth uniform sphere P has mass 0.3 kg . Another smooth uniform sphere Q , with the same radius as P , has mass 0.2 kg .

The spheres are moving on a smooth horizontal surface when they collide obliquely. Immediately before the collision the velocity of P is $(4\mathbf{i} + 2\mathbf{j})\text{ m s}^{-1}$ and the velocity of Q is $(-3\mathbf{i} + \mathbf{j})\text{ m s}^{-1}$.

At the instant of collision, the line joining the centres of the spheres is parallel to $\mathbf{i}$.

The kinetic energy of Q immediately after the collision is half the kinetic energy of Q immediately before the collision.

(a) Find

- (i) the velocity of P immediately after the collision,
- (ii) the velocity of Q immediately after the collision,
- (iii) the coefficient of restitution between P and Q ,
carefully justifying your answers.

(11)

- (b) Find the size of the angle through which the direction of motion of P is deflected by the collision.

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 14 marks)



6. A light elastic string with natural length l and modulus of elasticity kmg has one end attached to a fixed point A on a rough inclined plane. The other end of the string is attached to a package of mass m .

The plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

The package is initially held at A . The package is then projected with speed $\sqrt{6gl}$ up a line of greatest slope of the plane and first comes to rest at the point B , where $AB = 3l$.

The coefficient of friction between the package and the plane is $\frac{1}{4}$

By modelling the package as a particle,

- (a) show that $k = \frac{15}{26}$ (6)
- (b) find the acceleration of the package at the instant it starts to move back down the plane from the point B . (5)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 11 marks)



7.

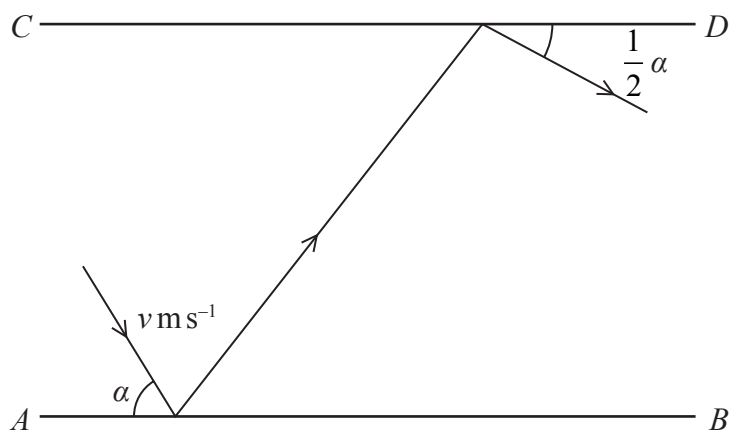


Figure 2

Figure 2 represents the plan view of part of a horizontal floor, where AB and CD represent fixed vertical walls, with AB parallel to CD .

A small ball is projected along the floor towards wall AB . Immediately before hitting wall AB , the ball is moving with speed $v \text{ m s}^{-1}$ at an angle α to AB , where $0 < \alpha < \frac{\pi}{2}$

The ball hits wall AB and then hits wall CD .

After the impact with wall CD , the ball is moving at angle $\frac{1}{2}\alpha$ to CD .

The coefficient of restitution between the ball and wall AB is $\frac{2}{3}$

The coefficient of restitution between the ball and wall CD is also $\frac{2}{3}$

The floor and the walls are modelled as being smooth. The ball is modelled as a particle.

(a) Show that $\tan\left(\frac{1}{2}\alpha\right) = \frac{1}{3}$ (7)

(b) Find the percentage of the initial kinetic energy of the ball that is lost as a result of the two impacts. (4)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.



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Question 7 continued

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(Total for Question 7 is 11 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel		Centre Number	Candidate Number
Level 3 GCE			
Thursday 20 June 2019			
Morning (Time: 1 hour 30 minutes)		Paper Reference 9FM0/3C	
Further Mathematics			
Advanced			
Paper 3C: Further Mechanics 1			
You must have: Mathematical Formulae and Statistical Tables (Green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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(b) Find the value of U .

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Question 1 continued

Handwriting practice area with 30 horizontal lines.

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Question 1 continued

Lined area for writing the answer to Question 1.

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Question 1 continued

Handwriting practice area with 28 horizontal lines.

(Total for Question 1 is 8 marks)



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2. A small box is projected with speed 7 m s^{-1} from a point O on a fixed rough inclined plane. The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$. The box moves up a line of greatest slope of the plane and comes to instantaneous rest at the point A . The coefficient of friction between the box and the plane is $\frac{1}{4}$. In a model of the motion, the box is modelled as a particle.

- (a) Show that, after coming to rest at A , the box immediately slides back down the plane. (2)

The speed of the box at the instant when it returns to O is $V \text{ m s}^{-1}$.

Given that $OA = \frac{25}{8}$ m,

- (b) use the work-energy principle to find the value of V . (4)

- (c) Suggest one way in which the model can be refined to make it more realistic. (1)

[illegible]

Question 2 continued

Lined area for writing the answer to Question 2 continued.

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Question 2 continued

Lined area for writing the answer to Question 2.

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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 7 marks)



3. A particle of mass 0.5 kg is moving with velocity $(-\mathbf{i} + 2\mathbf{j})\text{ ms}^{-1}$ when it receives an impulse $\mathbf{I}\text{ N s}$. As a result of the impulse, the kinetic energy of the particle increases by 12 J .

Given that $\mathbf{I}$ acts in the direction of $(2\mathbf{i} - \mathbf{j})$, find $\mathbf{I}$.

(7)

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 7 marks)



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4. Two smooth spheres, A and B , of the same radius, have masses $2m$ and $3m$ respectively. The spheres are at rest on a smooth horizontal plane. Sphere A is projected towards B with speed u and collides directly with B . The coefficient of restitution between the spheres is e , where $e > \frac{2}{3}$

- (a) Find, in terms of u and e ,

- the speed of A immediately after the collision,
- the speed of B immediately after the collision.

(7)

- (b) Describe the direction of motion of A immediately after the collision, justifying your answer.

(1)

Given that $e = \frac{5}{6}$

- (c) find the total kinetic energy lost in the collision between A and B .

(4)



Question 4 continued

Handwriting practice area with 20 horizontal lines.

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Question 4 continued

Handwriting practice area with 30 horizontal lines.

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 12 marks)



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Question 5 continued

Lined area for writing the answer to Question 5.

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Question 5 continued

Handwriting practice area with 30 horizontal lines.

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Question 5 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 5 is 15 marks)



6. Two fixed points, A and B , lie on a horizontal ceiling with $AB = 6a$. A light elastic string of modulus of elasticity $\frac{5mg}{3}$ has one end attached to A and the other end attached to B . A particle P of mass $4m$ is attached to the midpoint of the string and P hangs in equilibrium at a distance $4a$ below AB .

(a) Show that the natural length of the string is $4a$.

(5)

The particle P is now held at the midpoint of AB and released from rest.

(b) Find the maximum speed of P as it falls.

(6)

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Question 6 continued

Lined area for writing the answer to Question 6.

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Question 6 continued

Handwriting practice area with 30 horizontal lines.

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Question 6 continued

Handwriting practice area with 30 horizontal lines.

(Total for Question 6 is 11 marks)



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7. A small ball is projected with speed 14 ms^{-1} from a point O on the ground. The ball is projected at an angle α to the ground, where $\tan \alpha = \frac{3}{4}$. The ball bounces on the ground for the first time at the point A_1 . The coefficient of restitution between the ball and the ground is $\frac{1}{2}$. The ball is modelled as a particle moving freely under gravity from O to A_1 and between bounces. The ground is modelled as a smooth horizontal plane.
- (a) Find the size of the angle between the direction of motion of the ball and the ground immediately after the ball bounces on the ground at A_1
- (b) Explain how, in your calculation, you have used the fact that the ball is moving freely under gravity from O to A_1

The ball bounces on the ground for the second time at the point A_2 .

- (c) Find the total time taken by the ball to travel from O to A_2 (4)

The ball bounces on the ground for the n th time at the point A_n

Immediately after the ball bounces at A_n , the angle between the direction of motion of the ball and the ground is ϕ .

- (d) Find, in terms of n only, an expression for $\tan \phi$. (3)
- (e) Describe, according to the model, the subsequent motion of the ball after it has bounced on the ground at A_2 (1)

Given instead that the coefficient of restitution between the ball and the ground is 0

- (f) describe fully the motion of the ball from the instant when it is projected from O . (2)



Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Lined area for writing the answer to Question 7.

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Question 7 continued

Handwriting practice area with 30 horizontal lines.

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Question 7 continued

Lined area for writing answers.

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(Total for Question 7 is 15 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname		Other names	
Pearson Edexcel Level 3 GCE		Centre Number	Candidate Number
Mock Paper Set 1			
(Time: 1 hour 30 minutes)		Paper Reference 9FM0/3C	
Further Mathematics Advanced Paper 3C: Further Mechanics 1			
You must have: Mathematical Formulae and Statistical Tables (green), calculator			Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. A lorry of mass 2500 kg pulls a trailer of mass 500 kg along a straight horizontal road. The trailer is attached to the lorry by a tow bar that is horizontal and parallel to the direction of motion. The resistance to the motion of the lorry is 300 N and the resistance to the motion of the trailer is 120 N. The tow bar is modelled as a light rod.

At the instant when the speed of the lorry is 20 m s^{-1} , the lorry is accelerating, the tension in the tow bar is 200 N and the power being developed by the engine of the lorry is $P \text{ kW}$.

- (a) Find the value of P .

(6)

- (b) Explain how you have used the modelling assumption that the tow bar is inextensible in your solution to part (a).

(1)



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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 7 marks)



2. Two smooth uniform spheres P and Q , of equal radius, are both moving with speed u on a smooth horizontal plane. The spheres are moving in opposite directions along the same straight line when they collide directly. Sphere P has mass $2m$ and sphere Q has mass $3m$. The coefficient of restitution between the spheres is e .

Prove that if $e > \frac{1}{4}$ then the direction of motion of Q is reversed by the collision.

(8)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 8 marks)



3. A small ball has mass 0.6 kg . The ball is projected vertically upwards with speed 22.4 m s^{-1} from a point that is 1.5 m above horizontal ground. The ball is modelled as a particle and air resistance is ignored. The ground is soft and the ball sinks 2.5 cm into the ground before coming to rest. The resistive force exerted by the ground on the ball is modelled as a constant force of magnitude F newtons.

Find, using **only** the work-energy principle, the value of F .

(4)



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(Total for Question 3 is 4 marks)



4. Particle A , of mass m , is moving with speed u on a smooth horizontal plane towards an identical particle B , which is at rest on the plane. Particle A collides directly with particle B . The coefficient of restitution between the particles is e .

Given that 42% of the total initial kinetic energy of the two particles is lost in the collision,

- (a) find the value of e .

(10)

If instead $e = 0$,

- (b) describe, as fully as possible, the motion of the particles after the collision.

(2)



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Question 4 continued

Lined area for writing the answer to Question 4.



Question 4 continued

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Question 4 continued

Lined area for writing answers.

(Total for Question 4 is 12 marks)



5. A small ball of mass 0.5 kg is moving on a smooth horizontal plane with velocity $(4\mathbf{i} - \mathbf{j}) \text{ m s}^{-1}$ when it strikes a fixed vertical wall. Immediately after the impact the velocity of the ball is $(2\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The ball is modelled as a particle and the wall is modelled as a smooth plane surface.
- (a) Find the magnitude of the impulse of the wall on the ball in the impact. (4)
- (b) Find the loss in kinetic energy of the ball due to its impact with the wall. (3)
- (c) Find the coefficient of restitution between the ball and the wall. (5)
- (d) Verify that the component of the momentum of the ball, parallel to the line of intersection of the wall and the horizontal plane, is unchanged by the impact. (2)
- (e) State which modelling assumption ensures that the component of the momentum of the ball, parallel to the line of intersection of the wall and the horizontal plane, is unchanged by the impact. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 15 marks)



6. A light elastic spring, of natural length l and modulus of elasticity mg , has one end attached to a fixed point A on a rough horizontal table. The other end of the spring is attached to a particle P of mass m . The particle P is held on the table at a distance $2l$ from A . The coefficient of friction between P and the table is $\frac{1}{4}$

The particle is released from rest.

- (a) Find the distance travelled by P before it first comes to rest. (6)
- (b) Explain why, when P first comes to rest, it is only at rest for an instant. (3)
- (c) Find where P comes to rest for the second time. (5)
- (d) Describe what happens to P next, explaining your answer. (2)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 16 marks)



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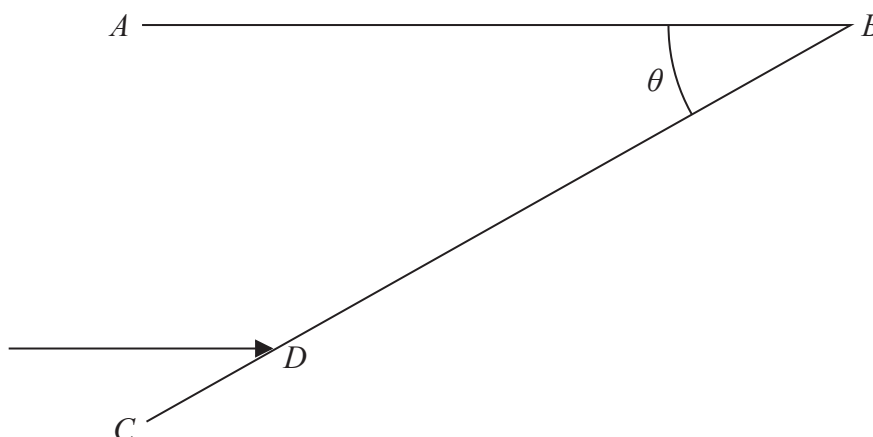


Figure 1

Figure 1 shows a plan view of two smooth fixed vertical walls, AB and BC , which are fixed to smooth horizontal ground. The angle between the walls is θ , where θ is an acute angle. A particle P is moving in a straight line on the ground and hits the wall BC at the point D .

Immediately before P hits the wall BC , P is moving in a direction that is parallel to the wall AB , as shown in Figure 1. Particle P bounces off BC and, after one impact with AB , P then hits BC for a second time. The coefficient of restitution between P and each wall is e .

Given that when P hits BC for a second time, P is moving in a direction that is perpendicular to BC ,

(a) show that

$$e(e + 2) \tan^2 \theta = 1 \quad (10)$$

(b) Deduce that, for the motion to be possible, the angle between the walls must be at least 30° .

(3)



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Question 7 continued

Lined area for writing the answer to Question 7.



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Question 7 continued

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Question 7 continued

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(Total for Question 7 is 13 marks)

TOTAL FOR PAPER IS 75 MARKS



Please check the examination details below before entering your candidate information

Candidate surname

Other names

**Pearson Edexcel
Level 3 GCE**

Centre Number

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Candidate Number

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Mock Paper Set 2

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/3C**

Further Mathematics

Advanced

Paper 3C: Further Mechanics 1

You must have:

Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

1. A car of mass 700 kg is moving along a straight horizontal road. At the instant when the speed of the car is $v\text{ m s}^{-1}$, the resistance to the motion of the car is modelled as a force of magnitude $(200 + 10v)\text{ N}$.

The engine of the car is working at a constant rate of 12 kW .

At the instant when the speed of the car is $w\text{ m s}^{-1}$, the acceleration of the car is 0.6 m s^{-2}

Find the value of w .

(5)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 5 marks)



2. A particle P of mass 2 kg is moving in a straight line on a smooth horizontal surface.

When the speed of P is 3 m s^{-1} , P receives an impulse $\mathbf{I}$ Ns.

Immediately after receiving the impulse, the speed of P is 9 m s^{-1} . As a result of the impulse, the direction of motion of P is deflected through an angle of 45° .

- Find
- (i) the magnitude of $\mathbf{I}$ Ns,
 - (ii) the size of the angle between the direction of $\mathbf{I}$ Ns and the direction of motion of P immediately before P receives the impulse.

(6)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



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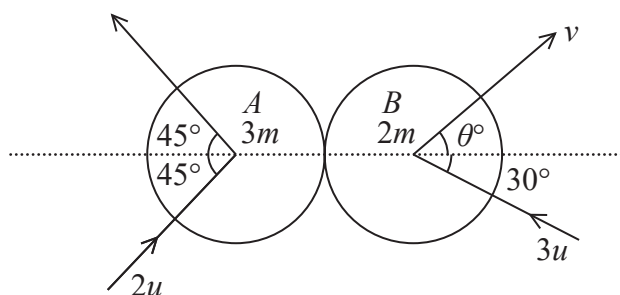


Figure 1

A smooth uniform sphere A has mass $3m$. Another smooth uniform sphere B , with the same radius as A , has mass $2m$.

The spheres are moving on a smooth horizontal surface when they collide obliquely.

Immediately before the collision

- the speed of A is $2u$ and the speed of B is $3u$
- sphere A is moving at 45° to the line of centres
- sphere B is moving at 30° to the line of centres

Immediately after the collision

- the speed of B is v
- sphere B is moving at θ° to the line of centres as shown in Figure 1.

The coefficient of restitution between A and B is e .

Given that the direction of motion of A is deflected through 90° by the collision,

- (a) find
- v in terms of u
 - the value of θ

(6)

- (b) Find the value of e .

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.



Question 3 continued

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Question 3 continued

Lined area for writing answers.

(Total for Question 3 is 9 marks)



4.



Figure 2

A boy cycles along a cycle route. The route passes through two points A and B . The boy descends from A , cycles along horizontal ground and then ascends to B . The point A is 15 m above the horizontal ground and the point B is 9 m above the horizontal ground.

At the instant the boy passes point A , his speed is 5 m s^{-1} .

At the instant the boy passes point B , his speed is 7 m s^{-1} , as shown in Figure 2.

The distance from A to B along the cycle route is 800 m. The resistance to the motion of the boy and his cycle from non-gravitational forces is modelled as a force of constant magnitude 12 N. The total mass of the boy and his cycle is 75 kg.

(a) Find the work done by the boy as he cycles from A to B .

(5)

At the instant the boy passes point B , he is moving horizontally, working at a rate of 140 W and accelerating at $k \text{ m s}^{-2}$.

(b) Find the value of k .

(3)



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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 8 marks)



5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A small ball of mass 1.5 kg is moving on a horizontal plane when it strikes a fixed smooth vertical wall. Immediately before the impact the velocity of the ball is $8\mathbf{i}\text{ ms}^{-1}$. Immediately after the impact the velocity of the ball is $\lambda(\mathbf{i} + 2\mathbf{j})\text{ ms}^{-1}$, where λ is a positive constant.

The coefficient of restitution between the ball and the wall is e .

The kinetic energy lost by the ball when it strikes the wall is 33 J .

(a) Find the impulse received by the ball when it strikes the wall.

(5)

(b) Find the value of e .

(6)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 11 marks)



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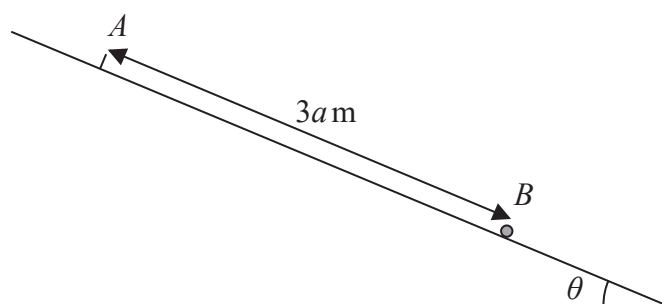


Figure 3

[In this question, use $g = 10 \text{ m s}^{-2}$]

A light elastic string has natural length a metres and modulus of elasticity λ newtons. A particle P of mass 2 kg is attached to one end of the string. The other end of the string is attached to a fixed point A on a rough inclined plane. The plane is inclined at angle θ

to the horizontal, where $\tan \theta = \frac{3}{4}$

The point B on the plane lies below A on the line of greatest slope of the plane through A and $AB = 3a$ metres, as shown in Figure 3.

The particle P is held at B and then released from rest. The particle first comes to instantaneous rest at A .

The coefficient of friction between P and the plane is $\frac{1}{4}$

- (a) Show that $\lambda = 24$ (5)
- (b) Find the magnitude of the acceleration of P at the instant it is released from B . (5)
- (c) Explain why the answer to part (b) is the greatest value of the magnitude of the acceleration of P as P moves from B to A . (1)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 11 marks)



7. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

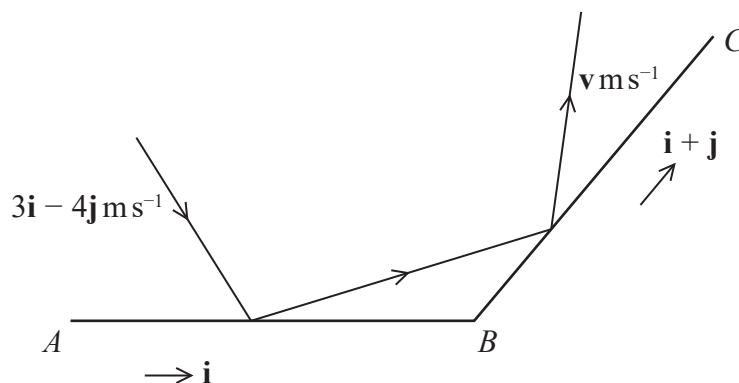


Figure 4

Figure 4 represents the plan view of part of a horizontal floor, where AB and BC are fixed vertical walls. The floor and the walls are modelled as being smooth.

The vector $\vec{AB}$ is in the direction of $\mathbf{i}$. The vector $\vec{BC}$ is in the direction of $(\mathbf{i} + \mathbf{j})$.

A ball is projected along the floor towards the wall AB so that, immediately before hitting the wall AB , the velocity of the ball is $(3\mathbf{i} - 4\mathbf{j})\text{ms}^{-1}$. The ball hits the wall AB and then

hits the wall BC . The coefficient of restitution between the ball and each wall is $\frac{1}{4}$

Immediately after hitting the wall BC , the velocity of the ball is $\mathbf{v}\text{ms}^{-1}$.

By modelling the ball as a particle,

- (a) find $\mathbf{v}$ in the form $(a\mathbf{i} + b\mathbf{j})$. (8)

The total angle through which the path of the ball has been deflected as a result of the two collisions is θ°

- (b) Find the value of θ to the nearest whole number. (3)



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Question 7 continued

Lined area for writing the answer to Question 7.



Question 7 continued

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Question 7 continued

Lined area for writing the answer to Question 7.

(Total for Question 7 is 11 marks)



8. Two particles, A and B , have masses $2m$ and $3m$ respectively. The particles are moving in the same direction along the same straight line on a smooth horizontal surface when they collide directly. Immediately after the collision, the speed of A is u , the speed of B is $\frac{8}{5}u$ and the particles are both moving in the same direction.

After the collision with A , particle B hits a fixed smooth vertical wall which is perpendicular to the direction of motion of B . Particle B rebounds from the wall so that there is a second collision between A and B .

The first collision between A and B occurs at a distance d from the wall. The second collision between A and B occurs at a distance x from the wall.

Given that the coefficient of restitution between B and the wall is $\frac{1}{2}$

- (a) find x in terms of d .

(5)

Given also that the coefficient of restitution between the particles is $\frac{1}{3}$

- (b) show that there will not be a third collision between A and B .

(9)



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Question 8 continued

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Question 8 continued

Lined area for writing the answer to Question 8.



Question 8 continued

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(Total for Question 8 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS



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Level 3 GCE

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Further Mathematics

Advanced
Further Mathematics Option 1
Paper 3: Further Mechanics 1
Further Mathematics Option 2
Paper 4: Further Mechanics 1

Sample Assessment Material for first teaching September 2017

Time: 1 hour 30 minutes

Paper Reference

9FM0/3C
9FM0/4C

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 8 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Answer ALL questions. Write your answers in the spaces provided.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

1. A particle P of mass 0.5 kg is moving with velocity $(4\mathbf{i} + \mathbf{j}) \text{ m s}^{-1}$ when it receives an impulse $(2\mathbf{i} - \mathbf{j}) \text{ N s}$.

Show that the kinetic energy gained by P as a result of the impulse is 12 J .

(6)

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2. A parcel of mass 5 kg is projected with speed 8 m s^{-1} up a line of greatest slope of a fixed rough inclined ramp.

The ramp is inclined at angle α to the horizontal, where $\sin \alpha = \frac{1}{7}$

The parcel is projected from the point A on the ramp and comes to instantaneous rest at the point B on the ramp, where $AB = 14 \text{ m}$.

The coefficient of friction between the parcel and the ramp is μ .

In a model of the parcel's motion, the parcel is treated as a particle.

- (a) Use the work-energy principle to find the value of μ .

(5)

- (b) Suggest one way in which the model could be refined to make it more realistic.

(1)

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3. A particle of mass m kg lies on a smooth horizontal surface.

Initially the particle is at rest at a point O between two fixed parallel vertical walls.

The point O is equidistant from the two walls and the walls are 4 m apart.

At time $t = 0$ the particle is projected from O with speed u m s⁻¹ in a direction perpendicular to the walls.

The coefficient of restitution between the particle and each wall is $\frac{3}{4}$

The magnitude of the impulse on the particle due to the first impact with a wall is λmu N s.

- (a) Find the value of λ .

(3)

The particle returns to O , having bounced off each wall once, at time $t = 7$ seconds.

- (b) Find the value of u .

(5)

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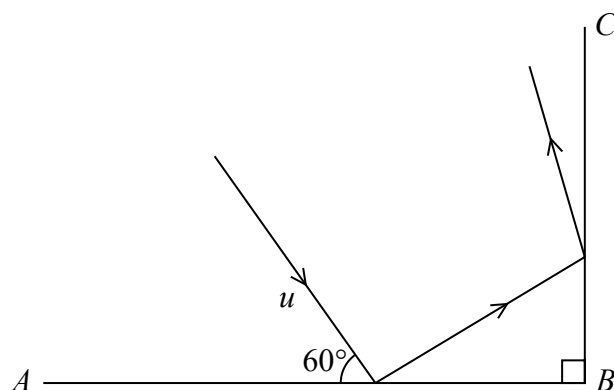
**Figure 1**

Figure 1 represents the plan view of part of a horizontal floor, where AB and BC are perpendicular vertical walls.

The floor and the walls are modelled as smooth.

A ball is projected along the floor towards AB with speed $u \text{ m s}^{-1}$ on a path at an angle of 60° to AB . The ball hits AB and then hits BC .

The ball is modelled as a particle.

The coefficient of restitution between the ball and wall AB is $\frac{1}{\sqrt{3}}$

The coefficient of restitution between the ball and wall BC is $\sqrt{\frac{2}{5}}$

- (a) Show that, using this model, the final kinetic energy of the ball is 35% of the initial kinetic energy of the ball.

(8)

- (b) In reality the floor and the walls may not be smooth. What effect will the model have had on the calculation of the percentage of kinetic energy remaining?

(1)

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Question 4 continued

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5. A car of mass 600 kg is moving along a straight horizontal road.

At the instant when the speed of the car is $v \text{ m s}^{-1}$, the resistance to the motion of the car is modelled as a force of magnitude $(200 + 2v) \text{ N}$.

The engine of the car is working at a constant rate of 12 kW.

- (a) Find the acceleration of the car at the instant when $v = 20$

(4)

Later on the car is moving up a straight road inclined at an angle θ to the horizontal,

where $\sin \theta = \frac{1}{14}$

At the instant when the speed of the car is $v \text{ m s}^{-1}$, the resistance to the motion of the car from non-gravitational forces is modelled as a force of magnitude $(200 + 2v) \text{ N}$.

The engine is again working at a constant rate of 12 kW.

At the instant when the car has speed $w \text{ m s}^{-1}$, the car is decelerating at 0.05 m s^{-2} .

- (b) Find the value of w .

(5)

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6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane.]

A smooth uniform sphere A has mass $2m$ kg and another smooth uniform sphere B , with the same radius as A , has mass $3m$ kg.

The spheres are moving on a smooth horizontal plane when they collide obliquely.

Immediately before the collision the velocity of A is $(3\mathbf{i} + 3\mathbf{j})\text{ms}^{-1}$ and the velocity of B is $(-5\mathbf{i} + 2\mathbf{j})\text{ms}^{-1}$.

At the instant of collision, the line joining the centres of the spheres is parallel to $\mathbf{i}$.

The coefficient of restitution between the spheres is $\frac{1}{4}$

(a) Find the velocity of B immediately after the collision.

(7)

(b) Find, to the nearest degree, the size of the angle through which the direction of motion of B is deflected as a result of the collision.

(2)

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Question 6 continued

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7. A particle P of mass m is attached to one end of a light elastic string of natural length a and modulus of elasticity $3mg$.

The other end of the string is attached to a fixed point O on a ceiling.

The particle hangs freely in equilibrium at a distance d vertically below O .

- (a) Show that $d = \frac{4}{3}a$. (3)

The point A is vertically below O such that $OA = 2a$.

The particle is held at rest at A , then released and first comes to instantaneous rest at the point B .

- (b) Find, in terms of g , the acceleration of P immediately after it is released from rest. (3)

- (c) Find, in terms of g and a , the maximum speed attained by P as it moves from A to B . (5)

- (d) Find, in terms of a , the distance OB . (3)

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Question 7 continued

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8. A particle P of mass $2m$ and a particle Q of mass $5m$ are moving along the same straight line on a smooth horizontal plane.

They are moving in opposite directions towards each other and collide directly.

Immediately before the collision the speed of P is $2u$ and the speed of Q is u .

The direction of motion of Q is reversed by the collision.

The coefficient of restitution between P and Q is e .

- (a) Find the range of possible values of e .

(8)

Given that $e = \frac{1}{3}$

- (b) show that the kinetic energy lost in the collision is $\frac{40mu^2}{7}$.

(5)

- (c) Without doing any further calculation, state how the amount of kinetic energy lost in the collision would change if $e > \frac{1}{3}$

(1)

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Question 8 continued

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Question 8 continued

(Total for Question 8 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS

Please check the examination details below before entering your candidate information

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Other names

**Pearson Edexcel
Level 3 GCE**

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Specimen Paper

(Time: 1 hour 30 minutes)

Paper Reference **9FM0/3C**

**Further Mathematics
Advanced
Paper 3C: Further Mechanics 1**

You must have:

Mathematical Formulae and Statistical Tables, calculator

Total Marks

Candidates may use any calculator permitted by Pearson regulations. Calculators must not have the facility for algebraic manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 7 questions in this question paper. The total mark for this paper is 75.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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Question 1 continued

Lined area for writing answers.

(Total for Question 1 is 9 marks)



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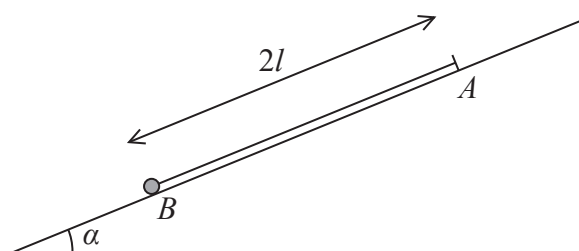


Figure 1

One end of a light elastic string, of natural length l and modulus of elasticity $\frac{3}{4}mg$, is attached to a particle of mass m . The other end of the string is attached to a fixed point A on a rough inclined plane. The plane is inclined at angle α to the horizontal, where

$$\tan \alpha = \frac{5}{12}$$

Initially the particle is held at the point B on the plane, where $AB = 2l$ and B lies below A on the line of greatest slope through A , as shown in Figure 1.

The particle is released from rest at B and first comes to instantaneous rest at the point C , where C is between A and B and $AC = \frac{8}{5}l$.

Find the coefficient of friction between the particle and the plane.

(6)



Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 6 marks)



Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 9 marks)



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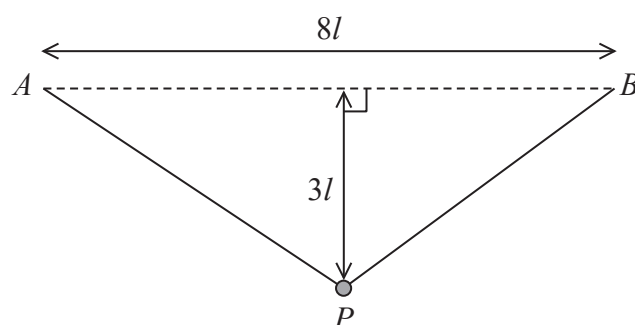


Figure 2

A light elastic string, of natural length $8l$ and modulus of elasticity kmg , has its ends attached to two points A and B , where $AB = 8l$ and AB is horizontal.

A pebble, P , of mass m is attached to the midpoint of the string. The pebble rests in equilibrium at a distance $3l$ vertically below AB , as shown in Figure 2. The pebble is modelled as a particle, and air resistance is modelled as negligible.

- (a) Show that $k = \frac{10}{3}$ (4)

The pebble is pulled vertically downwards from its equilibrium position until the total length of the string is $\frac{40}{3}l$. The pebble is released from rest.

- (b) Find the acceleration of P at the instant it is released from rest. (3)

At the instant the pebble crosses the line AB , the pebble has speed v .

- (c) Find v . (3)

In an experiment, when the natural length of the string was 2 m , it was found that the speed of P at the instant when it crossed the line AB was 1.5 m s^{-1} .

- (d) Considering the model, suggest a reason, other than air resistance, why the model and the experiment give different values. (1)



Question 4 continued

Lined area for writing the answer to Question 4.

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Question 4 continued

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Question 4 continued

Handwriting practice area with 20 horizontal lines.

(Total for Question 4 is 11 marks)



5. [In this question $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors in a horizontal plane]

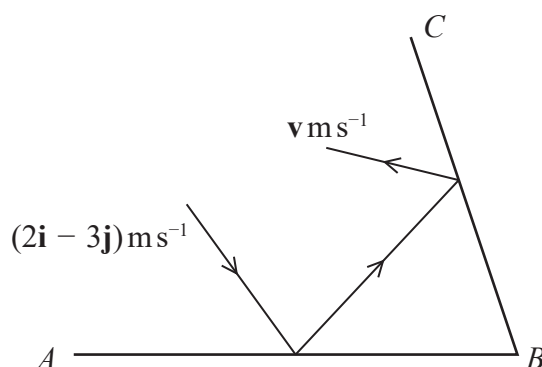


Figure 3

Figure 3 represents the plan view of part of a horizontal floor, where AB and BC represent fixed vertical walls. The direction of $\vec{AB}$ is in the direction of the vector $\mathbf{i}$ and the direction of $\vec{BC}$ is in the direction of the vector $(-\mathbf{i} + 3\mathbf{j})$.

A small ball is projected along the floor towards wall AB so that, immediately before hitting wall AB , the velocity of the ball is $(2\mathbf{i} - 3\mathbf{j})\text{ms}^{-1}$.

The ball hits wall AB and then hits wall BC .

The coefficient of restitution between the ball and wall AB is $\frac{1}{2}$

The coefficient of restitution between the ball and wall BC is $\frac{1}{3}$

The velocity of the ball immediately after hitting wall BC is $\mathbf{v}\text{ms}^{-1}$.

The floor and the walls are modelled as being smooth. The ball is modelled as a particle.

Show that $\mathbf{v} = \left(-\mathbf{i} + \frac{1}{2}\mathbf{j}\right)$.

(12)



Question 5 continued

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Question 5 continued

Handwriting practice area with 25 horizontal lines.

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Question 5 continued

Lined area for writing answers.

(Total for Question 5 is 12 marks)



6. A particle, P , of mass $4m$ is moving along a straight line on a smooth horizontal plane.

A particle, Q , of mass $3m$ is at rest on the plane on the same straight line.

Particle P collides directly with particle Q .

Immediately before the collision the speed of P is ku , where k is a constant.

Immediately after the collision the speed of P is u and the speed of Q is $\frac{3u}{2}$

The coefficient of restitution between P and Q is e .

- (a) (i) Show that there is only one possible value of k .

- (ii) State the value of k and the value of e .

(11)

- (b) Find the total kinetic energy lost in the collision between P and Q .

(3)



Question 6 continued

Handwriting practice area with 25 horizontal lines.

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Question 6 continued

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Question 6 continued

Lined area for writing answers.

(Total for Question 6 is 14 marks)



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Question 7 continued

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Question 7 continued

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Question 7 continued

Handwriting practice area with 20 horizontal lines.

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Question 7 continued

Lined area for writing answers.

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(Total for Question 7 is 14 marks)

TOTAL FOR PAPER IS 75 MARKS

